RAM KRISHNA UNIVERSITY

Admission Management System

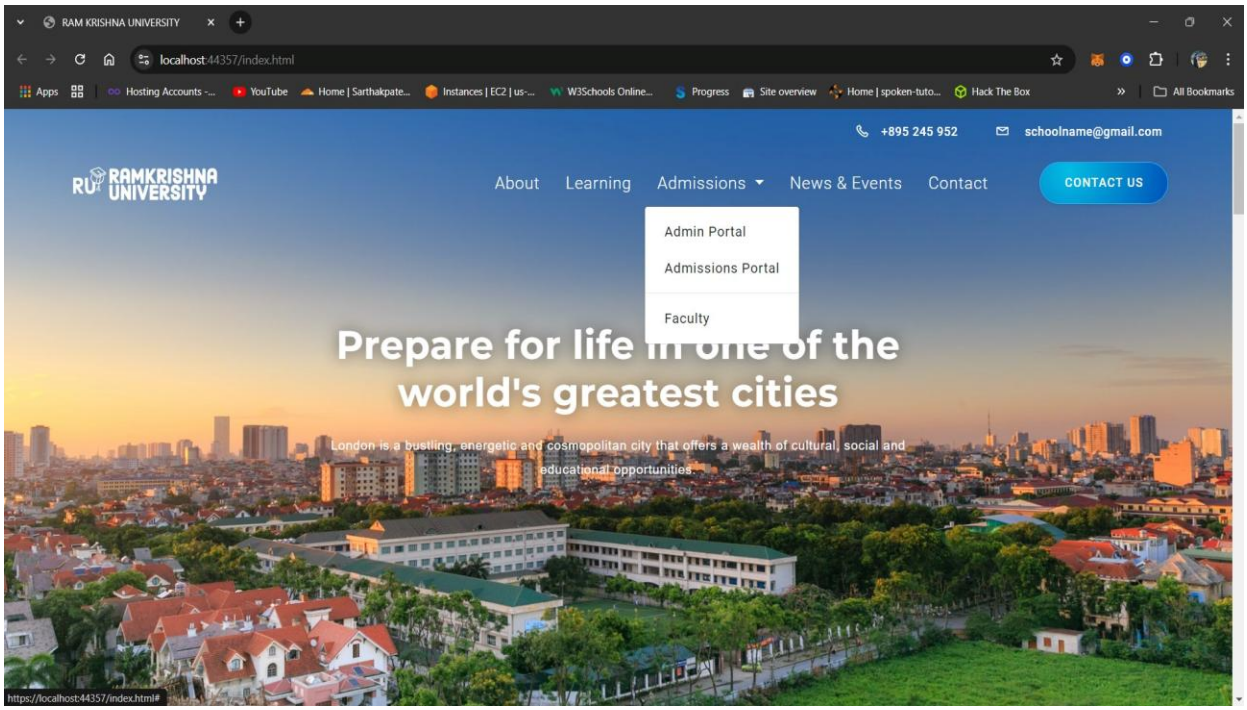

Figure: University Portal Homepage

Document Type	Project Documentation
Version	v1.0
Platform	ASP.NET Web Forms
Database	Microsoft SQL Server
Date	2024 – 2025

1. Executive Summary

The Ram Krishna University Admission Management System (RKU-AMS) is a comprehensive, role-based web application built on the ASP.NET Web Forms platform. It digitalises the entire university admission lifecycle — from initial student registration and email OTP verification, through faculty-level document checking, to final admin confirmation and merit-list generation.

The system enforces strict role-based access control (RBAC) with three distinct portals:

- Admin Portal — full system control, user management, department/faculty/course setup, and merit-list generation.
- Faculty Portal — document verification and final approval of student applications.
- Student (Admissions) Portal — self-registration with OTP-verified email, form submission, and status tracking.

Key Benefit

Eliminates manual paper-based admissions, provides real-time dashboards, enforces category-wise merit ranking, and delivers transparent verifiable workflows for all stakeholders.

2. Project Scope & Objectives

2.1 Scope

The system covers the following operational areas:

- Public-facing university website with navigation to all portals
- Secure admin authentication and multi-admin user management
- Hierarchical data setup — Departments, Colleges, Faculties, Courses, Fee Structures
- Student self-registration with email OTP verification
- Faculty document screening portal
- Automated merit-list creation by category and percentage threshold
- Excel export and printable merit lists
- Student confirmation workflow

2.2 Objectives

1. Provide a single, unified platform for all admission-related activities.
2. Enforce RBAC so each user sees only their authorised portal and data.
3. Automate merit ranking using configurable percentage cut-offs.

4. Send real-time email OTPs to validate applicant identity.
5. Enable faculty to review and verify student documents digitally.
6. Allow admins to export merit data to Excel for offline reporting.

3. Role-Based Access Control (RBAC)

RBAC is the cornerstone of the RKU-AMS security architecture. Every user is assigned exactly one role at account creation; the system redirects to the appropriate portal automatically upon login.

Role	Portal URL	Login Method	Key Capabilities
Admin	/Admin/admin_login.aspx	Username + Password	Full system control, manage users, generate merit lists
Faculty	/Faculty/Std_verification.aspx	Faculty credentials	Verify student documents, perform final verification
Student	/Admission_portal/Admission_Index.aspx	Email OTP	Register, submit form, track application status

3.1 Admin Role

Admins are the highest-privilege users. They manage the entire system, create other admin accounts, configure academic data, and oversee the full admission pipeline.

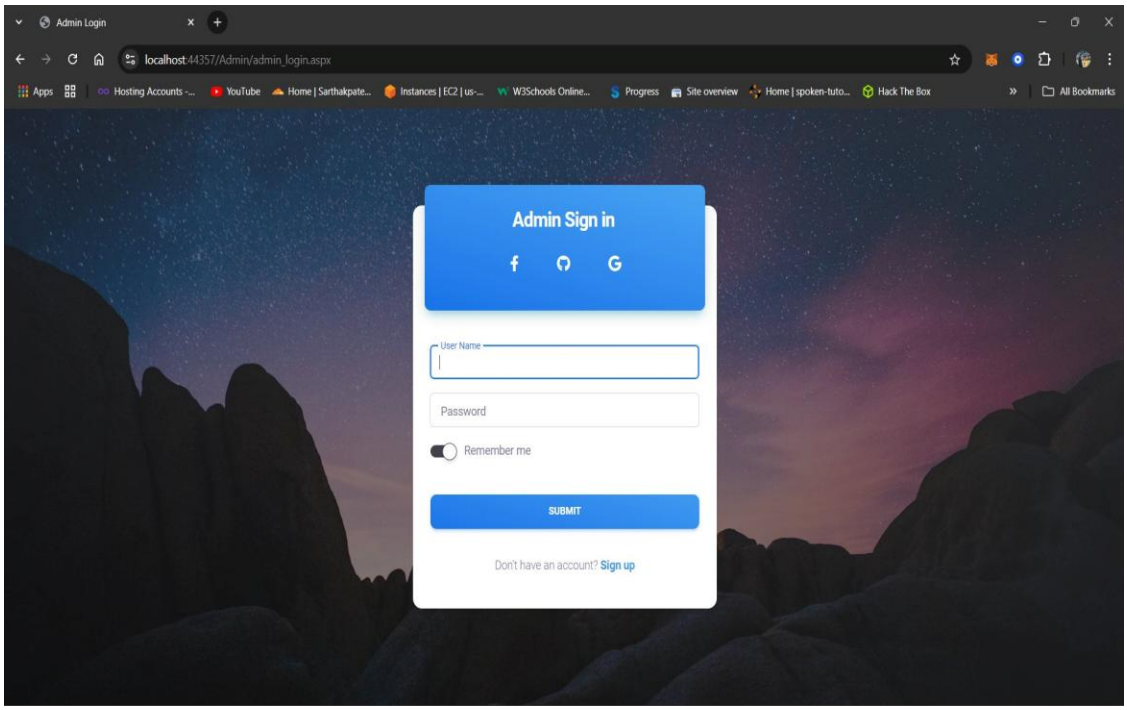

Figure: Admin Sign-In Page — Username/Password authentication

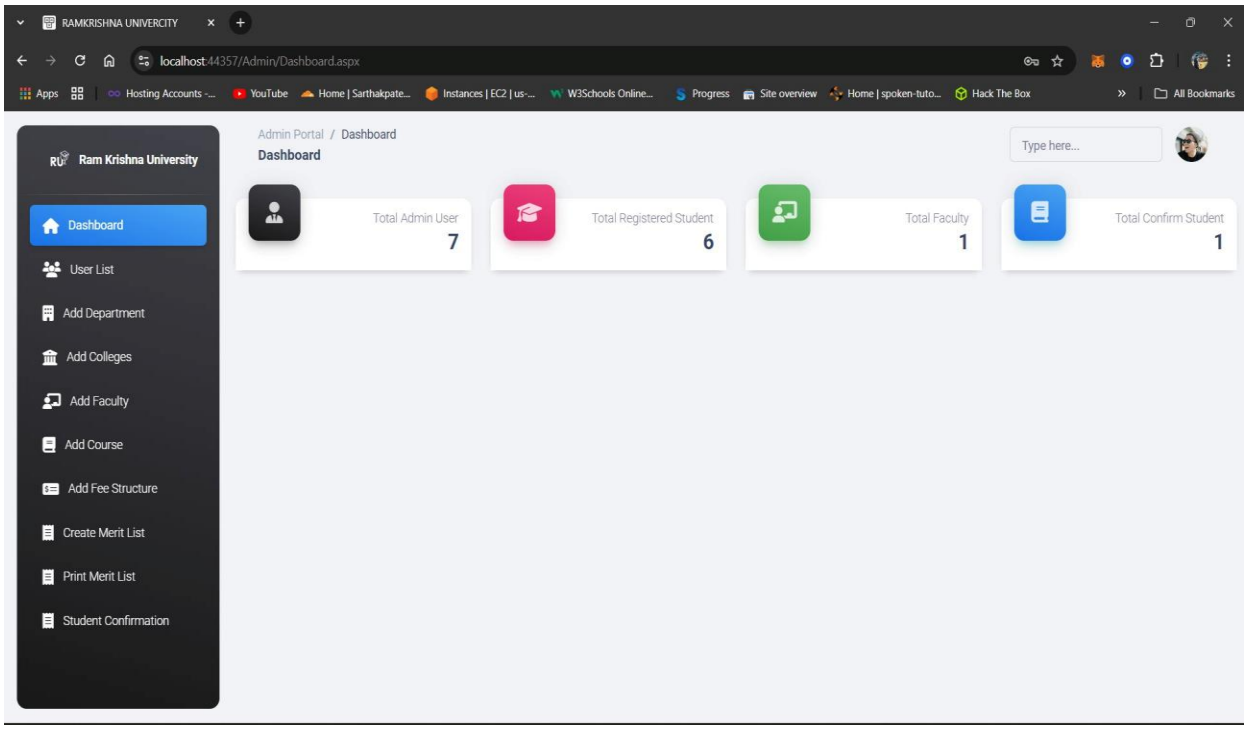

Figure: Admin Dashboard — Real-time KPI cards (Total Admin Users: 7 | Registered Students: 6 | Faculty: 1 | Confirmed: 1)

3.2 Faculty Role

Faculty members access a dedicated portal for verifying student-submitted documents. They perform both preliminary screening (Student Verification) and final approval (Final Verification).

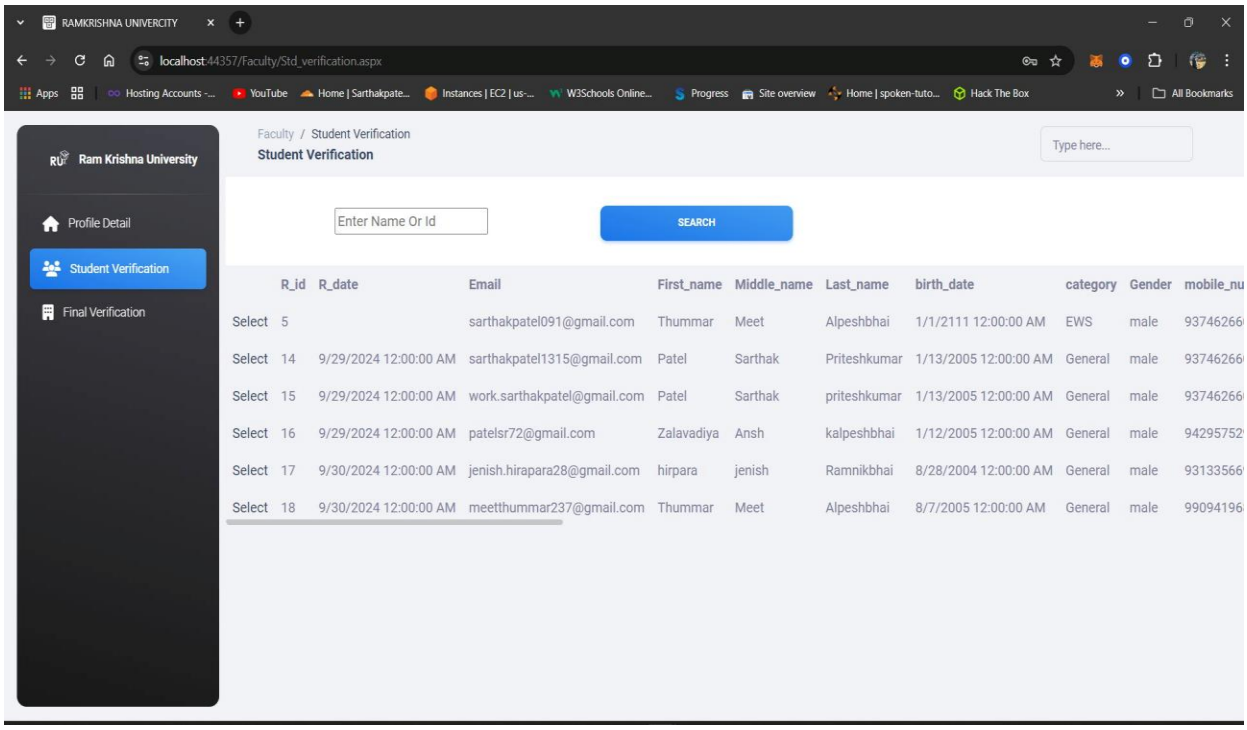

Figure: Faculty Portal — Student Verification listing with search capability

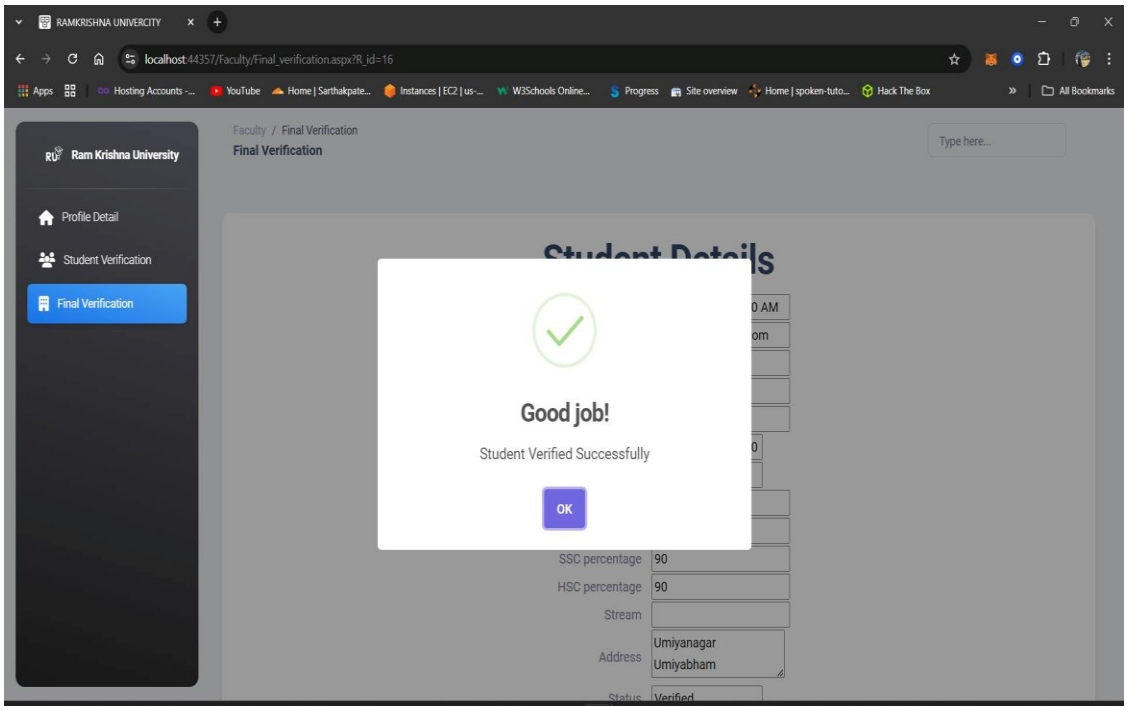

Figure: Faculty Portal — Final Verification success confirmation

3.3 Student (Admissions) Role

Students access the Admissions Portal. Before filling the registration form, they must verify their email address via a One-Time Password (OTP) delivered to their inbox. This ensures only valid applicants proceed.

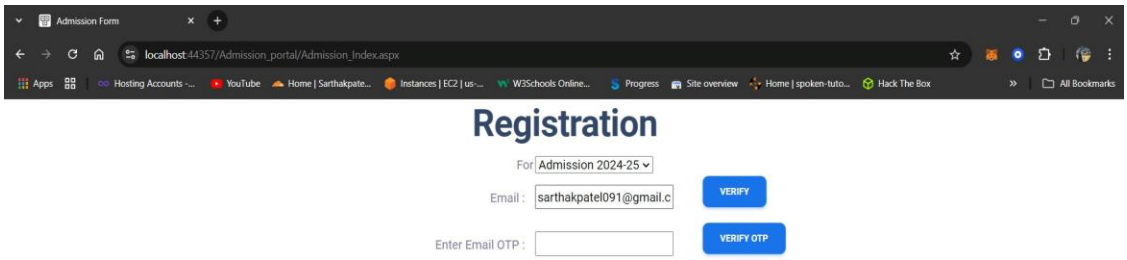

Figure: Admissions Portal — Email entry and OTP verification step

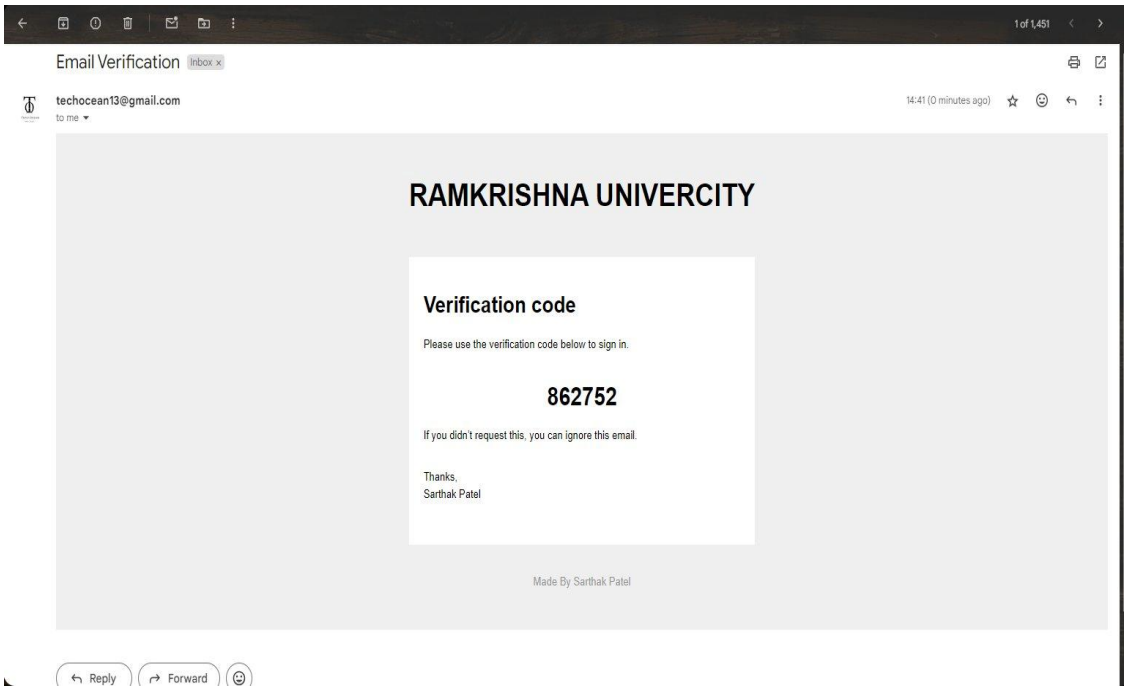

Figure: Actual OTP email received — 6-digit verification code from RAMKRISHNA UNIVERSITY

4. System Architecture

4.1 Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Bootstrap, JavaScript	Responsive UI across all portals
Backend	ASP.NET Web Forms (C#)	Business logic, session management, RBAC
Database	Microsoft SQL Server	Persistent storage for all entities
ORM / Data Access	ADO.NET / SqlDataSource	Database queries and stored procedures
Email Service	SMTP (Gmail)	OTP delivery and notifications
Reporting	Excel Export, Print-friendly views	Merit list output
Authentication	Custom session-based auth	Per-role portal routing

4.2 Portal URL Structure

Module	URL Pattern	Access Role
Home Page	/index.html	Public
Admin Login	/Admin/admin_login.aspx	Admin
Admin Dashboard	/Admin/Dashboard.aspx	Admin
User List	/Admin/user_list.aspx	Admin
Create Merit List	/Admin/create_merit_list.aspx	Admin
Print Merit List	/Admin/Print_merit_list.aspx	Admin
Student Verification	/Faculty/Std_verification.aspx	Faculty
Final Verification	/Faculty/Final_verification.aspx?R_id=XX	Faculty
Admissions Index	/Admission_portal/Admission_Index.aspx	Student (Public)

5. Module Descriptions

5.1 Public Homepage & Navigation

The public-facing homepage of Ram Krishna University serves as the entry point. The navigation bar contains a dropdown under 'Admissions' with three links:

- Admin Portal — directs administrators to the login page
- Admissions Portal — directs prospective students to the registration page
- Faculty — directs faculty members to the verification portal

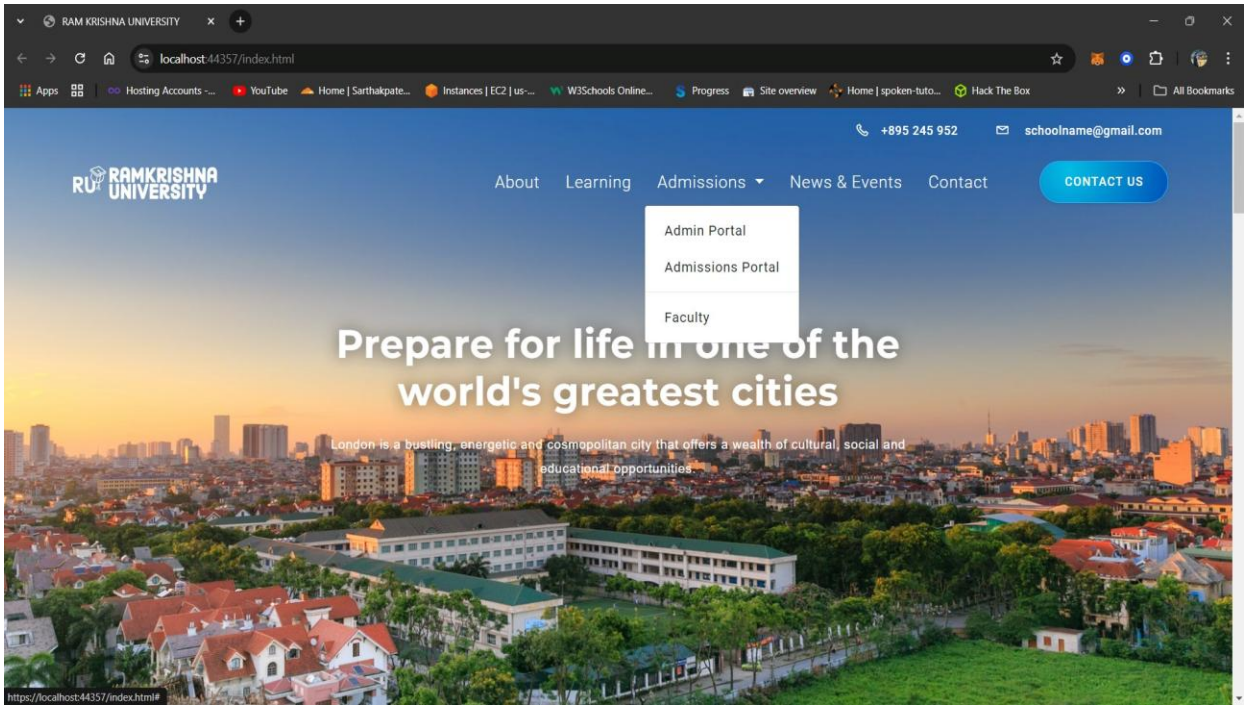

Figure: Homepage with Admissions dropdown showing portal navigation links

5.2 Admin Portal Modules

5.2.1 Admin Login

Secured login page at /Admin/admin_login.aspx with username-and-password authentication. Features include a 'Remember me' toggle, social provider icon display (Facebook, GitHub, Google), and a 'Don't have an account? Sign up' link for new admin registration.

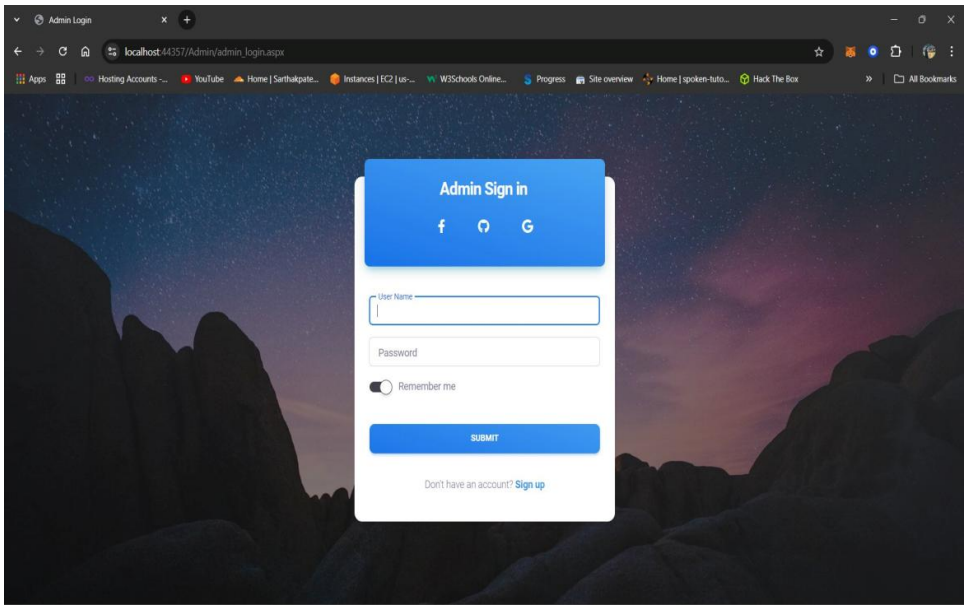

Figure: Admin Sign-In form with credential fields

5.2.2 Admin Dashboard

Upon successful login, admins land on a KPI dashboard that provides a real-time summary of system health:

- Total Admin Users — count of registered admin accounts
- Total Registered Students — cumulative applicant count
- Total Faculty — number of faculty members in the system
- Total Confirmed Students — students whose admissions have been finalised

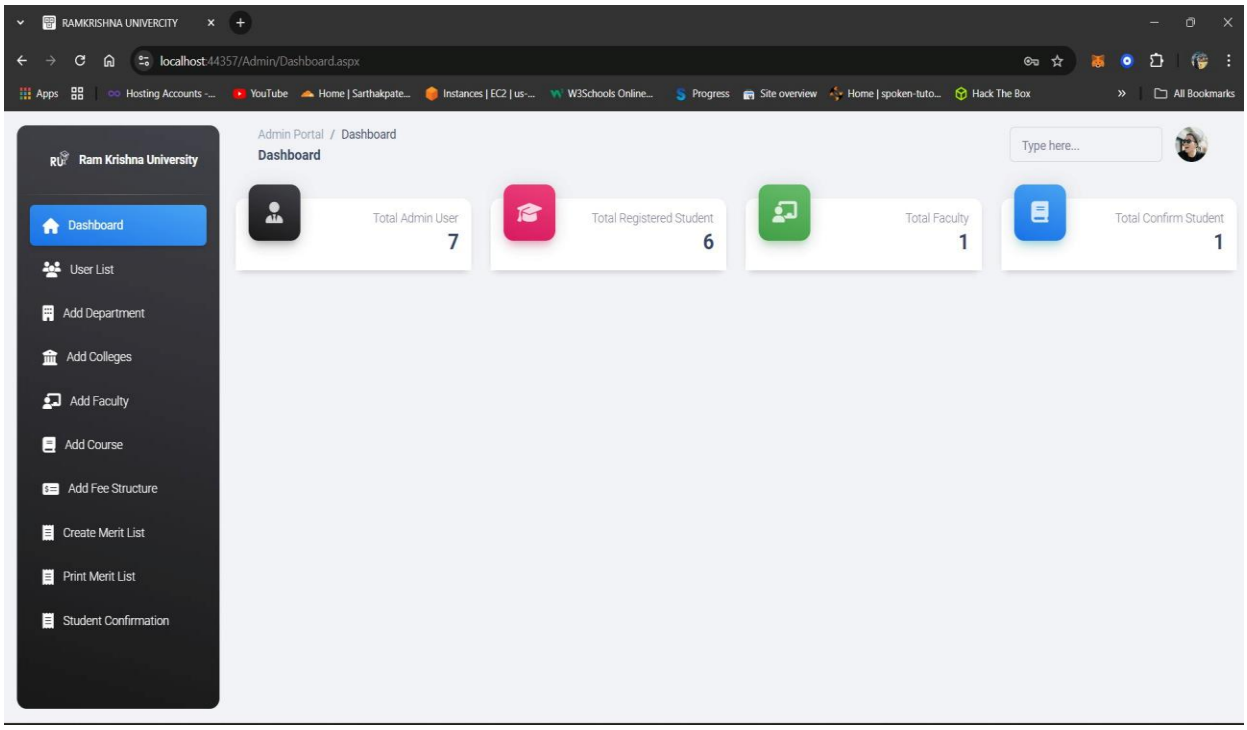

Figure: Admin Dashboard with four real-time KPI metric cards

5.2.3 User List & User Management

Admins can create, update, and delete admin-level user accounts. The form captures:

- User Name
- Password and Re-Type Password (client-side match validation)

The 'View User' grid below the form displays all existing users with their Admin_id, User_name, hashed Password, Role, and profile image path. Each row has Delete and Select action buttons.

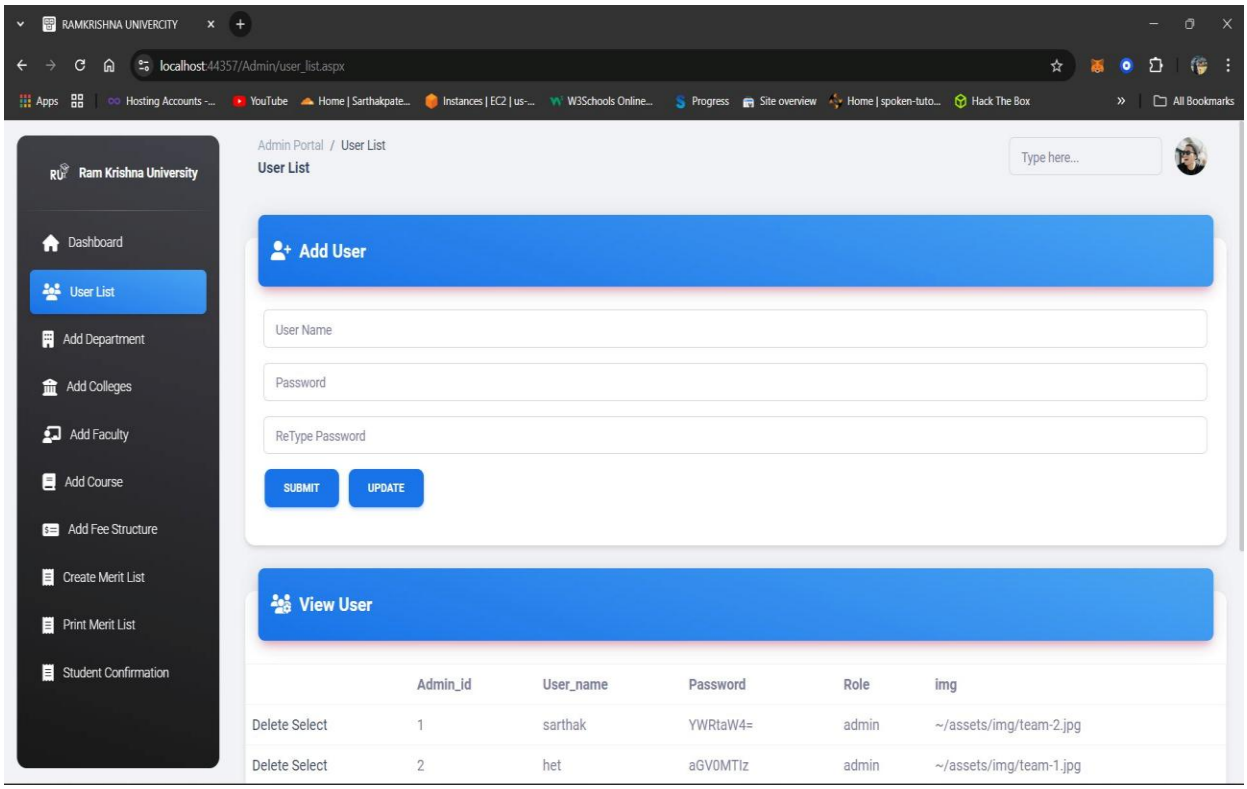

Figure: User List module — Add User form and existing user records grid

5.2.4 Add Department / Add Colleges / Add Faculty / Add Course / Add Fee Structure

These five modules follow the same create-read-update-delete (CRUD) pattern. Admins build the academic hierarchy — Departments contain Colleges, Colleges offer Courses, each with an associated Fee Structure. Faculty members are then assigned to the appropriate department.

5.2.5 Create Merit List

One of the most critical admin functions. The admin selects a student category (e.g., General, EWS, OBC) and enters a minimum percentage threshold. Clicking DISPLAY populates the Merit List table with all applicants in that category who meet or exceed the threshold. The admin then clicks CREATE to formally generate and persist the merit list.

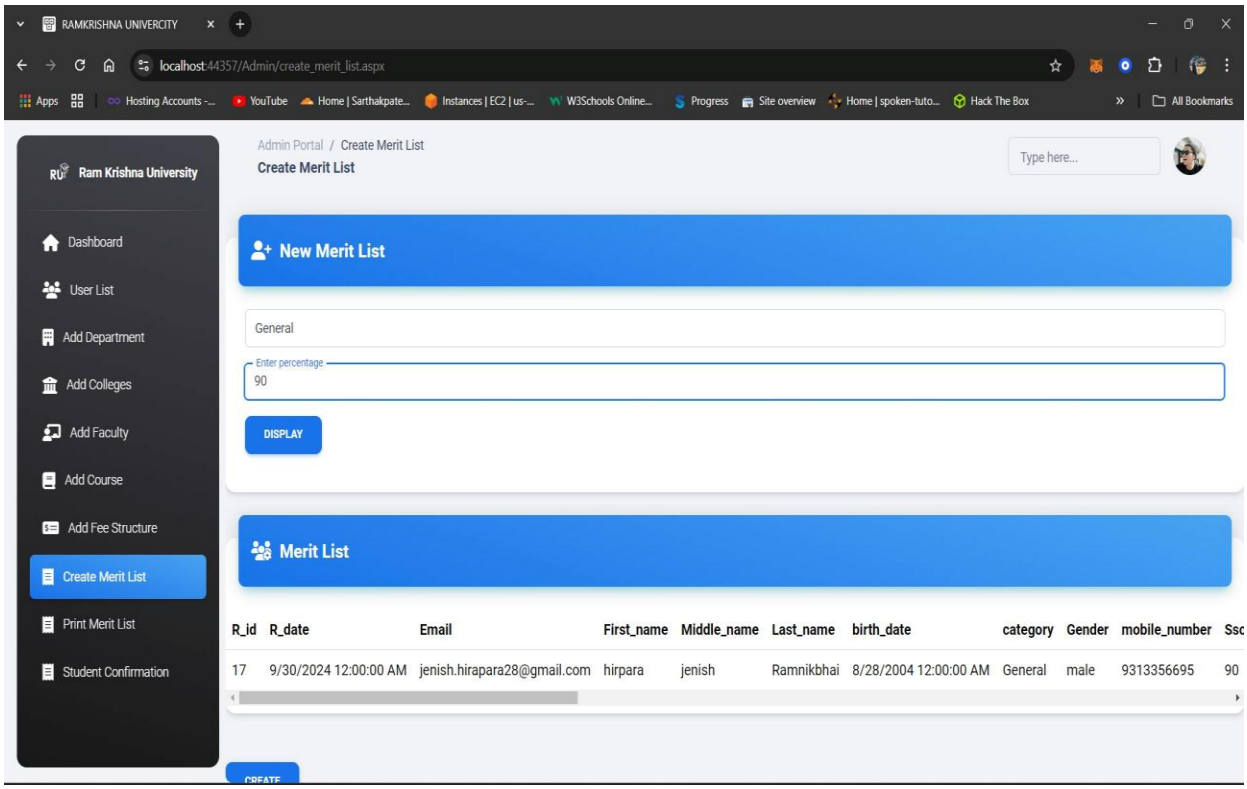

Figure: Create Merit List — Category filter set to 'General', threshold at 90%, student Jenish Hirpara Ramnikbhai displayed

5.2.6 Print Merit List

This module displays all persisted merit lists with their M_id, category, and percentage (pr). Selecting a merit list populates the detailed Merit List grid. Two export actions are available:

- PRINT — opens the browser print dialog for a formatted hardcopy
- EXPORT EXCEL — downloads the merit list as an .xlsx spreadsheet for offline use

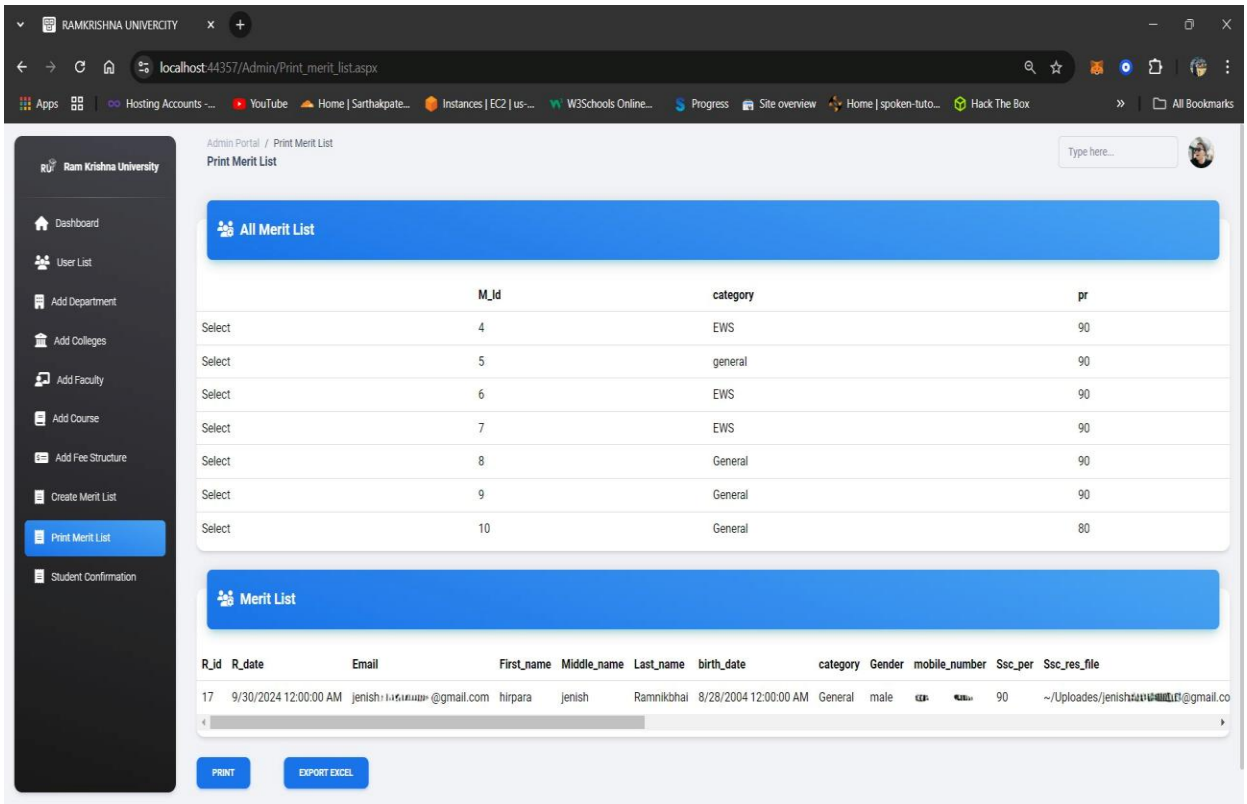

Figure: Print Merit List — all merit lists on the left, detailed student data on the right with Print and Export Excel

5.2.7 Student Confirmation

After merit-list creation and faculty verification, the admin performs a final confirmation step that officially registers the student for the admission year. Confirmed students contribute to the 'Total Confirm Student' count on the dashboard.

5.3 Faculty Portal Modules

5.3.1 Student Verification

Faculty can view all submitted applications in a searchable grid. Columns displayed include R_id, R_date, Email, First_name, Middle_name, Last_name, birth_date, category, Gender, and mobile_number. A free-text search field allows faculty to locate specific students by name or ID. Clicking 'Select' on any row advances to Final Verification.

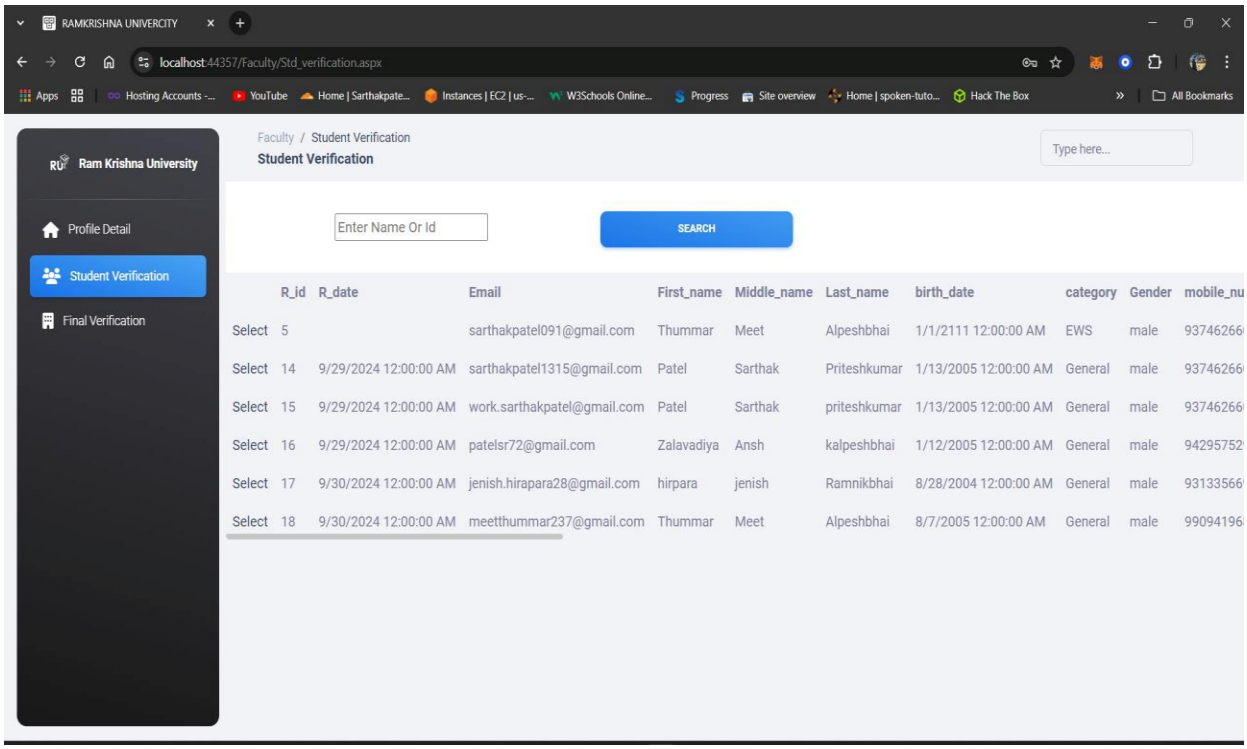

Figure: Faculty Student Verification grid with search and six student records

5.3.2 Final Verification

When a faculty member selects a student, they are taken to /Faculty/Final_verification.aspx?R_id=XX where they review the full student detail form — personal information, SSC percentage, HSC percentage, stream, address, and current status. After review, clicking Verify triggers a SweetAlert2 success modal confirming 'Student Verified Successfully'. The student's status is updated to 'Verified' in the database.

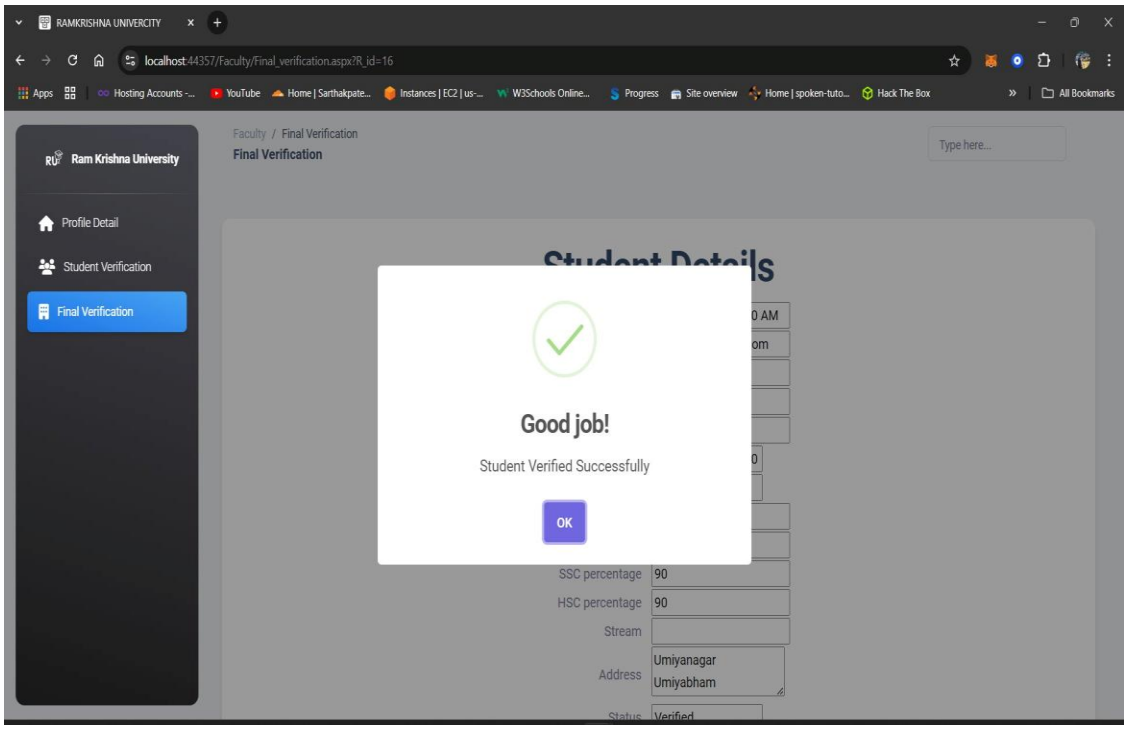

Figure: Faculty Final Verification — success modal after verifying student R_id 16

5.4 Student (Admissions) Portal Modules

5.4.1 Email OTP Verification

The admissions flow starts at /Admission_portal/Admission_Index.aspx. The student selects the admission year (e.g., 'Admission 2024-25'), enters their email address, and clicks VERIFY. The backend generates a 6-digit OTP and emails it via SMTP. The student then enters the OTP and clicks VERIFY OTP. Only upon successful OTP validation does the full registration form become accessible — this prevents bot submissions and ensures applicant email ownership.

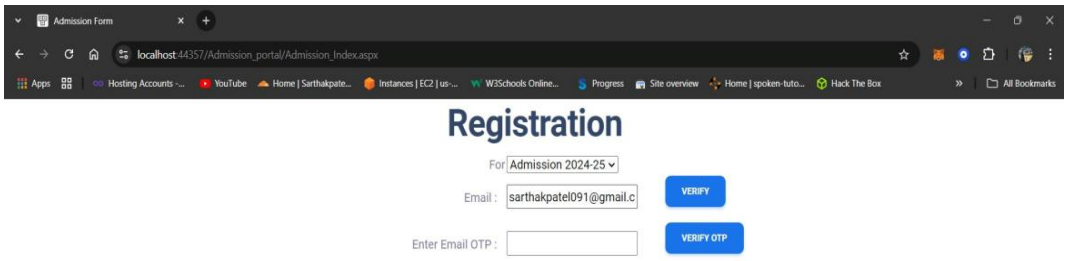

Figure: Admissions Portal — Email and OTP entry screen

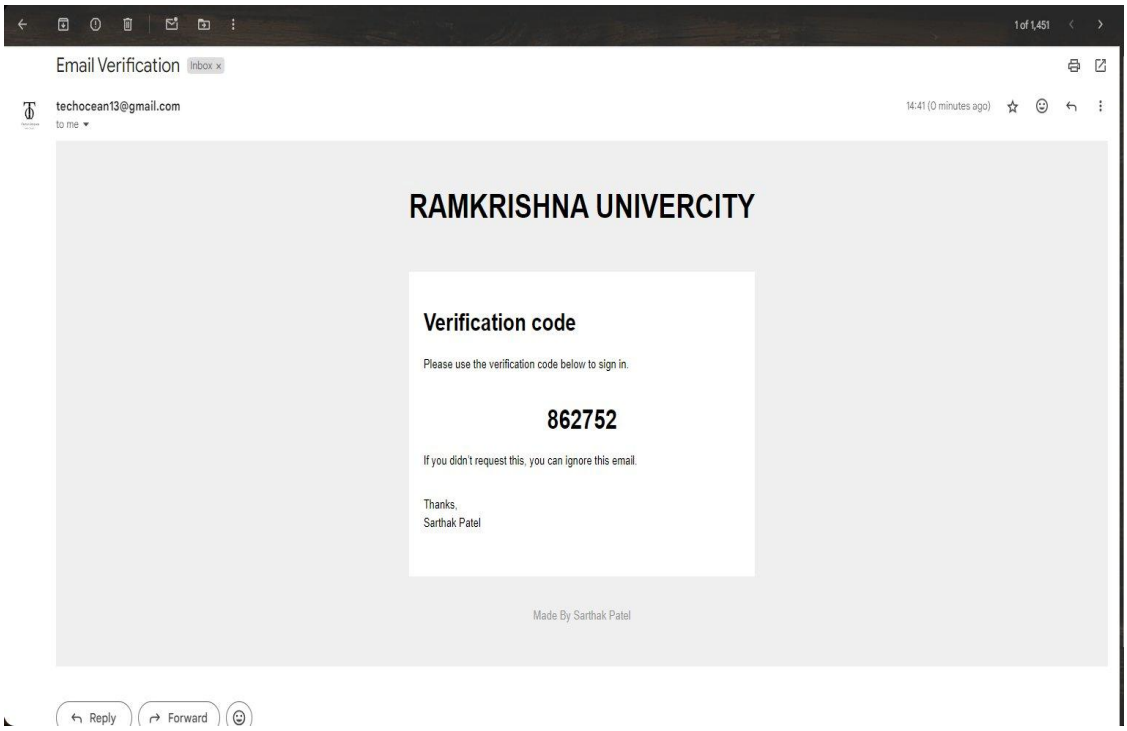

Figure: Actual verification email received with 6-digit OTP code 862752

5.4.2 Admission Form

After OTP verification, students complete the multi-section admission form capturing:

- Personal details — name, DOB, gender, category, mobile number
- Academic records — SSC percentage, HSC percentage, stream

- Address and correspondence details
- Document uploads — SSC result file (stored under ~/Uploades/<email>/)

6. Database Schema Overview

The SQL Server database contains the following primary tables:

Table	Key Columns	Description
Admin_Users	Admin_id, User_name, Password, Role, img	Stores all admin portal credentials and profile images
Registered_Students	R_id, R_date, Email, First_name, Middle_name, Last_name, birth_date, category, Gender, mobile_number, Ssc_per, Ssc_res_file, Status	All student registration records
Merit_Lists	M_id, category, pr (percentage)	Generated merit lists with category and threshold
Merit_Student_Link	M_id, R_id	Many-to-many: which students appear in which merit list
Departments	Dept_id, Dept_name	Academic departments
Colleges	College_id, College_name, Dept_id	Colleges linked to departments
Faculty	Faculty_id, Name, College_id, img	Faculty members assigned to colleges
Courses	Course_id, Course_name, College_id	Courses offered by colleges
Fee_Structure	Fee_id, Course_id, Amount	Fee amounts per course

7. Security Considerations

7.1 Authentication & Session Management

- Admin and Faculty portals use session-based authentication; unauthenticated requests redirect to the login page.
- Passwords in the Admin_Users table are stored as Base64-encoded strings (visible in screenshot: e.g., YWRtaW4=). Recommendation: upgrade to BCrypt hashing.
- OTP-based authentication for students prevents account hijacking without an email account.

7.2 Identified Security Concerns & Recommendations

Finding	Risk	Recommendation
Passwords stored as Base64	High — trivially reversible	Migrate to BCrypt / Argon2 hashing
OTP email from personal Gmail	Medium — deliverability, spoofing risk	Use a dedicated transactional email service (SendGrid, AWS SES)
R_id exposed in URL query string	Low-Medium — IDOR risk	Validate server-side that the faculty session owns the requested R_id
'Sign up' link on Admin login	Low — admin account creation publicly accessible	Remove or restrict behind an invite-only token flow
File upload to ~/Uploades/ folder	Medium — path traversal, file type abuse	Enforce MIME type validation and store outside webroot

8. End-to-End Admission Workflow

The following sequence describes the complete admission journey from a student's perspective through all system roles:

Step	Actor	Action	Outcome
1	Admin	Configure Departments, Colleges, Courses, Fee Structures	Academic hierarchy available for applicants
2	Student	Visit Admissions Portal, enter email, request OTP	OTP sent to registered email
3	Student	Enter OTP, verify email	Registration form unlocked
4	Student	Complete and submit admission form with documents	Record created in Registered_Students (Status: Pending)
5	Faculty	Log in to Faculty Portal, open Student Verification	List of pending applicants visible
6	Faculty	Select a student, review documents, click Verify	Student status updated to 'Verified'
7	Admin	Open Create Merit List, set category and % threshold	Eligible students displayed
8	Admin	Click CREATE to generate merit list	Merit list persisted in Merit_Lists table
9	Admin	Open Print Merit List, select list, click PRINT or EXPORT EXCEL	Merit list available as hardcopy or spreadsheet
10	Admin	Open Student Confirmation, confirm selected students	Status updated to Confirmed; dashboard count incremented

9. Non-Functional Requirements

Attribute	Requirement	Implementation Notes
Performance	Dashboard loads in < 2 seconds	Indexed database queries on R_id, category
Scalability	Support 1,000+ concurrent registrations	Connection pooling via ADO.NET; consider read replicas
Availability	99.5% uptime during admission season	Deploy on IIS with Application Request Routing
Security	HTTPS enforced, sessions expire after 30 min idle	SSL certificate on IIS; FormsAuthentication timeout
Usability	Mobile-responsive UI	Bootstrap CSS grid adapts to tablet/phone widths
Accessibility	Basic WCAG 2.1 AA compliance	Alt text on images, label-input associations
Audit Trail	Log all verification actions	SQL audit table with timestamp, Faculty_id, R_id

10. Future Enhancements

7. Student Self-Service Dashboard — let applicants log in to track their application status and download the admission letter.
8. Multi-round Admission — support multiple merit-list rounds with waitlist management.
9. SMS Notifications — send OTP and status updates via SMS alongside email.
10. Payment Gateway Integration — collect application fees online through Razorpay or PayU.
11. Role-based Analytics — per-department enrollment analytics visible to department heads.
12. API Layer — expose RESTful APIs so a mobile app can interface with the backend.
13. Document AI Verification — auto-validate SSC/HSC certificates using OCR before faculty review.
14. BCrypt Password Migration — replace Base64 with a modern hashing algorithm for admin credentials.

11. Glossary

Term	Definition
AMS	Admission Management System
RBAC	Role-Based Access Control — access rights assigned by role, not individual
OTP	One-Time Password — time-limited numeric code for identity verification
SSC	Secondary School Certificate (Class 10 board exam)
HSC	Higher Secondary Certificate (Class 12 board exam)
EWS	Economically Weaker Section — reservation category
Merit List	Ranked list of eligible applicants by category and percentage threshold
KPI	Key Performance Indicator — metrics shown on the admin dashboard
DXA	Document XML Attribute unit used in .docx page layout (1440 DXA = 1 inch)
SMTP	Simple Mail Transfer Protocol — used to send OTP verification emails

— End of Document — Ram Krishna University Admission Management System v1.0 — 2025 —